THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET D

Chapters 7 & 8: **Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The SI unit of power is: 1

(a) kilowatt-hour **(b)** watt **(c)** joule **(d)** newton

2. A stretched rubber band stores: 1

(a) kinetic energy **(b)** potential energy **(c)** light energy **(d)** no energy

3. A crowbar is a lever of the class in which: 1

(a) the fulcrum is between effort and load **(b)** there is no fulcrum

(c) the load is between fulcrum and effort **(d)** the effort is between fulcrum and load

4. Which has the same value as 1 J? 1

(a) 1 kg m s^{-2} **(b)** 1 kg m s^{-1} **(c)** 1 kg m^{-1} s^{-2} **(d)** 1 kg m^2 s^{-2}

5. 1 kilowatt equals: 1

(a) 10 W **(b)** 1000 W **(c)** 100 W **(d)** 10 000 W

6. Sound is produced by: 1

(a) steady objects **(b)** magnetic fields

(c) light waves **(d)** vibrating objects

7. The frequency of ultrasonic waves is: 1

(a) above 20 kHz **(b)** exactly 20 Hz

(c) between 20 Hz and 20 kHz **(d)** below 20 Hz

8. Pitch of a sound depends mainly on its: 1

(a) amplitude **(b)** speed **(c)** frequency **(d)** intensity

9. The minimum time gap needed to hear an echo separately is: 1

(a) 0.01 s **(b)** 0.1 s **(c)** 1 s **(d)** 0.5 s

10. If the frequency of a wave in a medium increases while the speed remains the same, the wavelength: 1

(a) stays the same **(b)** becomes infinite **(c)** increases **(d)** decreases

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What are the three parts of a lever? 2

12. State the principle of the lever. 2

13. What is echolocation? Name two animals using it. 2

14. What does the unit decibel measure? Give two typical values. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. State two kinds of energy transfer other than doing work. 3

16. A parking sensor emits ultrasound that is reflected from an obstacle 1.2 m away (v = 345 m s^{-1}). Find the time taken for the round trip. 3

17. A truck of mass 10 000 kg moving at 20 m s^{-1} enters a sand ramp. Find its initial kinetic energy and explain how it stops. 3

OR

Why can a perpetual-motion machine not exist? 3

18. Sound in steel travels at 5000 m s^{-1} and in air at 340 m s^{-1}. How much sooner does a sound travel through 340 m of steel than through air? 3

OR

Explain the working of the human ear in brief. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Explain why the total work cannot be reduced by a machine.
(ii) A man pushes a 100 kg load up a 5 m long ramp 1 m high (g = 10); find the effort without friction.
(iii) What is the work done? 5

OR

(i) Distinguish negative from positive work with two examples.
(ii) A scooter of mass 160 kg reaches speed v in the same time with and without a 40 kg passenger; find the ratio of fuel used. 5

20. (i) How is sound used for exploration beyond hearing? Give two applications.
(ii) Explain audio surveillance and the use of infrasound. 5

OR

Case study: Two friends stand 340 m apart along a steel fence. One strikes the fence; sound reaches the other through steel (5000 m s^{-1}) and air (340 m s^{-1}). 5
(i) Find the time through air.
(ii) Find the time through steel.
(iii) Find the time difference.
(iv) Can she hear two separate sounds?

END OF QUESTION PAPER — **Total: 40 Marks**